BASIC DATA EXPLORATION

ASSIGNMENT QUESTIONS & ANSWERS

MICROSOFT EXCEL • DATA ANALYTICS

Project Overview

This document combines the assignment questions from the Word file with the calculated results and formulas recorded in the supplied Excel workbook.

01 | Sum, Count, and Average

QUESTION Calculate the total price of all products in the dataset.

ANSWER The total price of all products is **$10,100**.

```
EXCEL FORMULA
=SUM(brand_Data[Price ($)])
```

QUESTION Determine how many products exist in the dataset.

ANSWER There are **34 products** in the dataset.

```
EXCEL FORMULA
=COUNTA(brand_Data[Product Name])
```

QUESTION Calculate the average price of the products.

ANSWER The average product price is **$297.06** (rounded to two decimal places).

```
EXCEL FORMULA
=AVERAGE(brand_Data[Price ($)])
```

02 | Minimum and Maximum Values

QUESTION Identify the minimum price among all products.

ANSWER The minimum product price is **$30**.

```
EXCEL FORMULA
=MIN(brand_Data[Price ($)])
```

QUESTION Identify the maximum price among all products.

ANSWER The maximum product price is **$1,000**.

```
EXCEL FORMULA
=MAX(brand_Data[Price ($)])
```

03 | Logical Function – IF

QUESTION Create a new column named Price Range and categorize each product: Price ≥ $500 as High Price; Price < $500 as Standard Price.

ANSWER The **Price Range** column uses the IF function. The sample result shown in the workbook is **High Price**.

```
EXCEL FORMULA
=IF([@[Price ($)]]>=500,"High Price","Standard Price")
```

04 | Conditional Functions – SUMIF and COUNTIF

QUESTION Use SUMIF to calculate the total price of products in the Electronics category.

ANSWER The total price of products in the **Electronics** category is **$8,050**.

```
EXCEL FORMULA
=SUMIF(brand_Data[Category],"Electronics",brand_Data[Price ($)])
```

QUESTION Use COUNTIF to determine the number of products with a price less than $100.

ANSWER **11 products** are priced below $100.

```
EXCEL FORMULA
=COUNTIF(brand_Data[Price ($)],"&lt;100")
```

05 | Text Formatting – LEFT, RIGHT, MID

QUESTION Create a Day column containing the first two characters of each Product ID using LEFT.

ANSWER For the sample Product ID, **28-JAN-US**, the Day value is **28**.

```
EXCEL FORMULA
=LEFT([@[Product ID]],2)
```

QUESTION Create a Country Code column containing the last two characters of each Product ID using RIGHT.

ANSWER For the sample Product ID, **28-JAN-US**, the Country Code is **US**.

```
EXCEL FORMULA
=RIGHT([@[Product ID]],2)
```

QUESTION Create a Month column containing characters 4–6 of each Product ID using MID.

ANSWER For the sample Product ID, **28-JAN-US**, the Month value is **JAN**.

```
EXCEL FORMULA
=MID([@[Product ID]],4,3)
```

Calculated Columns — All 34 Products

Results below are taken from the calculated results table in the supplied Excel workbook.

Product ID	Price Range	Day	Country Code	Month
28-JAN-US	High Price	28	US	JAN
15-FEB-US	Standard Price	15	US	FEB
03-MAR-US	Standard Price	03	US	MAR
11-APR-US	High Price	11	US	APR
22-MAY-US	Standard Price	22	US	MAY
07-JUN-UK	Standard Price	07	UK	JUN
19-JUL-UK	Standard Price	19	UK	JUL
23-AUG-UK	Standard Price	23	UK	AUG
05-SEP-UK	High Price	05	UK	SEP
14-OCT-UK	Standard Price	14	UK	OCT
17-JUN-IN	High Price	17	IN	JUN
25-NOV-AU	Standard Price	25	AU	NOV
08-DEC-DE	Standard Price	08	DE	DEC
18-FEB-CA	Standard Price	18	CA	FEB
16-APR-ES	Standard Price	16	ES	APR
21-AUG-CA	Standard Price	21	CA	AUG
20-AUG-CN	Standard Price	20	CN	AUG
27-JAN-IT	High Price	27	IT	JAN
01-MAR-UK	Standard Price	01	UK	MAR
14-AUG-US	Standard Price	14	US	AUG
14-MAY-RU	High Price	14	RU	MAY
09-JAN-CA	Standard Price	09	CA	JAN
19-JUL-BR	Standard Price	19	BR	JUL
21-AUG-CA	Standard Price	21	CA	AUG
29-SEP-CA	High Price	29	CA	SEP
03-JUN-CA	Standard Price	03	CA	JUN
11-JUL-CA	Standard Price	11	CA	JUL
16-APR-ES	Standard Price	16	ES	APR
07-MAR-CA	Standard Price	07	CA	MAR
13-APR-CA	Standard Price	13	CA	APR
24-MAY-CA	Standard Price	24	CA	MAY
02-DEC-CA	Standard Price	02	CA	DEC
17-JUN-IN	High Price	17	IN	JUN
09-JUL-FR	Standard Price	09	FR	JUL

CONCLUSION

This assignment demonstrates basic data exploration in Excel using summary, logical, conditional, and text functions. It provides a foundation for more advanced data analytics tasks.